

ROSTOCK LAAGE, Germany

WMO: 101720

Lat: 53.9156N

Lon: 12.2789E

Elev: 59

StdP: 100.62

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.1	-13.3	1.2	-9.8	-10.4	1.6	-7.3	13.0	5.7	11.6	5.5	3.1	90	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.8	(c) 29.1	(d) 19.2	(e) 27.1	(f) 18.7	(g) 25.1	(h) 17.8	(i) 20.5	(j) 26.6	(k) 19.5	(l) 25.2	(m) 18.5	(n) 23.6	(o) 4.2	(p) 80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.2	22.4	17.4	12.5	21.5	16.7	12.0	20.7	59.0	26.3	55.5	25.2	52.6	23.6	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.0	7.9	-13.6	32.8	3.8	1.9	-16.4	34.2	-18.6	35.3	-20.7	36.3	-23.5	37.7
			-13.8	22.6	3.8	0.9	-16.5	23.2	-18.8	23.8	-20.9	24.3	-23.6	24.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	0.5	1.5	3.8	8.5	12.5	15.7	18.1	18.0	14.3	9.7	5.0	1.8
	DBStd	7.26	4.58	4.33	3.84	3.82	3.51	3.07	2.99	2.88	2.86	3.45	3.49	4.46
	HDD10.0	1269	294	239	193	73	14	1	0	0	2	48	153	253
	HDD18.3	3434	552	472	449	296	184	91	42	42	126	268	400	511
	CDD10.0	962	0	0	2	27	91	171	251	249	130	38	3	0
	CDD18.3	86	0	0	0	0	3	12	34	32	4	0	0	0
	CDH23.3	837	0	0	0	8	44	139	326	281	38	1	0	0
	CDH26.7	211	0	0	0	1	7	36	88	74	4	0	0	0
(14) Wind	WSAvg	3.8	4.1	4.2	4.3	4.0	3.7	3.6	3.6	3.3	3.5	3.5	3.6	3.9
(15) Precipitation	PrecAvg	620	49	41	40	34	53	67	67	63	56	47	50	55
	PrecMax	797	95	88	101	71	104	136	287	145	160	87	117	120
	PrecMin	434	0	7	8	3	12	2	6	27	12	20	6	14
	PrecStd	90	26	17	21	17	24	29	51	29	32	17	24	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	12.1	16.7	24.3	27.5	30.2	31.9	31.2	27.1	20.8	14.6	12.0
		MCWB	9.3	9.1	10.8	15.3	18.3	20.2	20.5	19.9	18.2	16.0	12.6	10.2
	2%	DB	9.2	10.1	13.8	20.0	24.0	27.0	29.0	28.8	24.1	18.0	12.8	10.2
		MCWB	7.8	8.1	9.2	13.2	16.7	18.7	19.6	18.8	17.0	14.5	11.0	9.0
	5%	DB	7.9	8.8	11.2	17.9	21.8	24.2	27.0	26.5	21.8	16.1	11.2	8.9
		MCWB	6.7	7.1	8.1	11.7	15.1	17.3	19.0	18.5	16.0	13.4	9.7	7.8
	10%	DB	6.7	7.3	9.2	15.4	19.7	22.1	24.9	24.2	19.4	14.7	9.9	7.4
		MCWB	5.7	5.7	6.9	10.6	14.1	16.3	17.9	17.7	15.0	12.6	8.7	6.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	9.7	11.5	15.7	19.2	21.2	22.1	21.6	19.1	16.6	13.2	11.0
		MCDB	11.0	10.9	15.0	22.3	25.6	27.5	29.3	28.6	24.7	20.0	14.2	12.1
	2%	WB	8.0	8.4	9.8	13.8	17.5	19.6	20.6	20.1	17.7	14.9	11.3	9.2
		MCDB	8.9	9.8	12.4	19.2	22.6	25.6	26.8	25.8	22.4	17.3	12.6	10.0
	5%	WB	6.7	7.1	8.4	12.3	15.9	18.3	19.6	19.1	16.7	13.7	9.9	7.7
		MCDB	7.6	8.5	10.9	16.7	20.6	23.0	25.5	24.3	20.6	15.8	10.9	8.6
	10%	WB	5.5	5.7	7.2	10.9	14.4	16.9	18.5	18.2	15.6	12.8	8.7	6.4
		MCDB	6.4	7.1	9.1	14.8	18.6	21.1	23.5	23.0	19.0	14.5	9.6	7.2
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.3	7.1	9.7	10.4	9.9	9.8	9.9	8.7	7.0	4.9	4.2
		MCDBR	5.0	6.2	10.5	13.8	14.3	14.2	13.7	13.9	12.5	8.8	5.6	4.7
	5% WB	MCWBR	4.5	4.7	6.9	7.9	7.8	7.2	6.0	5.9	6.4	5.8	4.5	4.3
		MCDDBR	4.9	6.0	9.4	12.3	12.7	12.4	12.4	11.9	10.9	8.1	5.1	4.6
(40) Clear-Sky Solar Irradiance	taub		0.315	0.338	0.370	0.397	0.392	0.390	0.409	0.406	0.374	0.359	0.336	0.308
		taud	2.315	2.313	2.256	2.252	2.309	2.336	2.302	2.323	2.395	2.396	2.364	2.307
	Ebn at Noon		638	747	799	829	855	859	833	812	793	716	608	565
		Edh at Noon	59	81	105	120	120	118	120	111	92	74	56	49
(44) All-Sky Solar Radiation	RadAvg		0.55	1.19	2.42	4.10	5.17	5.49	5.10	4.22	3.06	1.62	0.68	0.39
	RadStd		0.07	0.18	0.25	0.53	0.60	0.41	0.66	0.32	0.31	0.22	0.08	0.03

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.50	N/A	N/A	N/A	+0.39	+0.54	-169	-239	+76	N/A
(47) Regional (52 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-164	N/A	N/A

Nomenclature: See separate page